

What Does Source Density Add to Models of Hydra Patterning and Regeneration?

Szymon Cygan · Anna Marciniak-Czochra · Finn Münnich · Moritz Mercker
Heidelberg University, Heidelberg, Germany

HEIDELBERG UNIVERSITY
LOGO PLACEHOLDER

01 | Biological motivation

HYDRA AS A PATTERNING PROBLEM

A Hydra must regenerate a single head, preserve its oral–aboral axis, and respond differently to tissue fragments with different origins.

HYDRA / WNT MOTIVATION
head regeneration and body-axis organization

One model must separate three phenomena: regeneration, pattern selection, and positional memory.

We ask which behaviours already follow from an activator–inhibitor system and which genuinely require an additional state variable.

02 | Start from the general model

PRODUCTION TERMS CHANGE THE PHASE PORTRAIT

$$\begin{aligned}\partial_t u &= D_u \Delta u + \frac{au^2}{v} - \mu_u u + p_u, \\ \partial_t v &= D_v \Delta v + bu^2 - \mu_v v + p_v.\end{aligned}$$

Here, u is the head-forming activator and v is the inhibitor. The three production terms are not interchangeable:

- p_v : basal inhibitor production; regularizes the resting state,
- p_u : external activator input,
- q_u : activator input inside inhibitor regulation.

Homogeneous equilibria satisfy the cubic

$$P(u) = \mu_u bu^3 - (a\mu_v + bp_u)u^2 + \mu_u p_v u - (\mu_v q_u + p_u p_v) = 0.$$

PHASE-PORTRAIT PLACEHOLDER
one versus three homogeneous states
nullclines + vector field + stability markers

1 state high activator production can remove the threshold.
3 states lower rest + unstable threshold + upper active state.
DII an outer stable kinetic state can become diffusion unstable.

From the next section onward: $p_u = q_u = 0$; only inhibitor production p_v is retained.

03 | Inhibitor production creates a threshold

MINIMAL REGENERATION MODEL

$$\begin{aligned}\partial_t u &= D_u \Delta u + \frac{au^2}{v} - \mu_u u, \\ \partial_t v &= D_v \Delta v + bu^2 - \mu_v v + p_v.\end{aligned}$$

The resting state is $E_0 = (0, p_v/\mu_v)$. Positive states exist when

$$\Delta_h = a^2 \mu_v^2 - 4 \mu_u^2 b p_v > 0,$$

with

$$u_{\pm} = \frac{a \mu_v \pm \sqrt{\Delta_h}}{2 \mu_u b}, \quad v_{\pm} = \frac{a}{\mu_u} u_{\pm}.$$

E_- is the activation threshold; E_+ is the candidate pattern-forming state.

Basal inhibitor production already gives a stable rest state, bistability, and a finite-amplitude trigger.

04 | Regeneration needs a finite wound

JULIA SIMULATION PLACEHOLDER
weak versus strong localized perturbation

A weak perturbation returns to rest. A sufficiently strong localized perturbation crosses E_- and produces a persistent activator peak. This threshold response does **not** require source density.

05 | Pattern selection depends on history

JULIA SIMULATION PLACEHOLDER
same final domain, two preparation histories
direct DII: many peaks | growth continuation: one peak

Starting near the diffusion-unstable state on a large domain can select several peaks. Starting on a small domain and increasing its size can preserve one established peak in the same final parameter regime.

Domain growth selects among coexisting patterns—but it does not remember where a tissue fragment came from.

06 | Grafting exposes missing information

ORIENTATION MATTERS

EXPERIMENTAL FIGURE PLACEHOLDER
Livshits et al. (K. Keren group)
oriented versus anti-oriented H2F grafts

Reversing the orientation of otherwise comparable tissue pieces changes the distribution of regeneration outcomes. Near the resting state, u and v alone need not distinguish an upper-body fragment from a lower-body fragment.

Which state variable records the original position and orientation of the grafted tissue?

07 | Prescribed $\rho(x)$ as tissue competence

$\rho(x)$: inherited positional information

PROFILE PLACEHOLDER
oriented and anti-oriented piecewise $\rho(x)$

A fixed source-density profile can encode tissue origin before the activator is present. It biases where a wound-triggered peak forms, following the source-gradient idea used in Hydra models (Suzuki & Takagi, 2011).

08 | Source-density model

AFTER THE AUDIT, KEEP ONLY p_v

$$\begin{aligned}\partial_t u &= D_u \Delta u + \rho \frac{au^2}{v} - \mu_u u, \\ \partial_t v &= D_v \Delta v + b \rho u^2 - \mu_v v + p_v, \\ \tau \partial_t \rho &= D_\rho \Delta \rho + u - \rho.\end{aligned}$$

The new variable has a narrow job description:

- $\rho(0, x)$ stores the grafted configuration,
- τ sets the lifetime of memory,
- D_ρ communicates positional information in space.

Source density is tested as a memory variable—not as the origin of basic pattern formation.

09 | Dynamic ρ gives metastable memory

FAST PATTERNING, SLOW REORGANIZATION

JULIA SIMULATION PLACEHOLDER
long-lived peak near the graft + peak position $x_{\max}(t)$

On the fast u – v time scale, the inherited ρ profile anchors a peak near the graft. As ρ relaxes, the peak can drift toward a position preferred by the activator–inhibitor subsystem.

The model can display positional memory for a long time without making the graft-selected peak truly stationary.

Working interpretation: metastability rather than a new asymptotically stable head position.

10 | Diffusion sets the spatial reach of memory

PARAMETER-SWEEP PLACEHOLDER
peak lifetime versus D_ρ and τ

- D_ρ **too small**: memory remains sharply localized.
- D_ρ **intermediate**: a usable positional gradient forms.
- D_ρ **too large**: spatial information is rapidly erased.

The relevant observable is a memory time, for example the first time $x_{\max}(t)$ leaves a prescribed neighbourhood of the graft.

Take-home message

- p_v **is enough** for a stable resting state, an activation threshold, and diffusion-driven patterning.
- Domain growth changes pattern selection** and can preserve one peak where several are also possible.
- ρ **adds a distinct capability**: inherited positional information for grafted tissue.
- Dynamic ρ naturally suggests metastability**: long-lived anchoring followed by slow reorganization.

Source density is not needed to make a head. It may be needed to remember where a head should form.

Outlook. A separate head-identity variable h is reserved for future work if stable organizer identity cannot be obtained from the (u, v, ρ) dynamics alone.